Financial Stance and Sentiment Classification: A Multi-Task and Layer-Wise Fine-Tuning Study on FOMC and Financial PhraseBank

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TABLE I: Project scope and credit claim for UNSW COMP6713 2026 T1 (155+ credits, 80 minimum).

Part	Component	Deliverable	Credits
A (Problem)	2 NLP problems × 5	Stance (hawkish / dovish / neutral) and sentiment (pos / neg / neu) classification	10
	2 text domains × 5	Central-bank communications (FOMC) and corporate financial news (PhraseBank)	10
B (Data)	2 datasets × 5	FOMC Hawkish–Dovish [5] and Financial PhraseBank [6]	10
	Domain lexicon	Loughran–McDonald financial sentiment dictionary [1]	10
C (Models)	Required baseline	TF-IDF + Logistic Regression	10
	3 alternative baselines	TF-IDF+SVM, TF-IDF trigrams+LR, TF-IDF+LM lexicon features	15
	3 pre-trained × ZS/FS	FinBERT [3] (zero + few-shot), BERT [2] (few), RoBERTa (few)	15
	FinBERT fine-tuning	Full fine-tune with class-weighted CE and linear schedule	20
	Extension 1 (novel)	Multi-task FinBERT (shared encoder, two heads) [7]	20
	Extension 2	BERT-base with LLRD + gradual unfreezing + label smoothing	15
D (Evaluation)	Quantitative	Accuracy, macro-F1, per-class F1 on held-out test sets	10
	Qualitative	Error analysis on real FOMC / PhraseBank sentences	5
	CLI	`predict.py` command-line inference	5
	Gradio demo	Two-tab interactive web UI (stance and sentiment)	10
	HF Hub release	Public checkpoint and demo card on Hugging Face Hub	5
		Total claimed	**155**

***Abstract*—We study two financial-NLP classification tasks on distinct text domains: monetary-policy *stance* (hawkish / dovish / neutral) on FOMC communications [5] and three-class *sentiment* on the Financial PhraseBank all-agree split [6]. Starting from a TF-IDF + logistic regression baseline, we build a ladder of twelve systems that spans linear classifiers, the Loughran–McDonald lexicon, zero-/few-shot probes over FinBERT / BERT / RoBERTa, full fine-tuning, and a multi-task FinBERT with a shared encoder and two task heads. Our multi-task FinBERT reaches 97.79% accuracy (0.9666 macro-F1) on sentiment and 64.92% accuracy (0.6384 macro-F1) on stance. A carefully fine-tuned *general-purpose* BERT-base with layer-wise learning-rate decay, gradual unfreezing and label smoothing *ties* the multi-task model (97.57% / 0.9670 sentiment; 65.12% / 0.6383 stance). Sentiment is near-saturated; stance–especially the dovish/hawkish boundary–remains the harder problem and is where domain pre-training and multi-task learning still pay off in low-data regimes.**

I. Introduction

Financial text analysis has become a de-facto downstream benchmark for domain-adapted language models [3, 4]. Two tasks dominate the practitioner literature. *Sentiment* classification on corporate news, exemplified by the Financial PhraseBank (FPB) [6], asks whether a sentence conveys a positive, negative or neutral outlook for an investor. *Stance* classification on central bank communications, framed by Shah et al. [5] as the Hawkish–Dovish problem, asks whether a sentence signals a tightening (*hawkish*), easing (*dovish*) or non-committal (*neutral*) monetary-policy posture. The two problems share surface vocabulary yet differ sharply: sentiment is strongly lexical ("record profit", "loss"), while stance depends on modal hedging and context ("some participants noted", "price pressures have eased").

This paper reports a controlled comparison of twelve systems on both tasks using the public `gtfintechlab` Hugging Face splits. Our contributions are:

- A *wide baseline ladder* TF-IDF with logistic regression, LinearSVC, trigrams, and hand-crafted Loughran–McDonald lexicon features establishing a strong non-neural floor.
- A zero-/few-shot study over three encoders (FinBERT, BERT-base, RoBERTa-base) that isolates the contribution of domain pre-training with almost no labelled data.
- A *multi-task FinBERT* with a shared encoder and two linear heads, trained by alternating one stance batch and one sentiment batch per iteration.

- A second extension that fine-tunes a *general-purpose* BERT-base with layer-wise learning-rate decay (LLRD), gradual unfreezing and label smoothing, which we find *ties* the multi-task domain model on both tasks.

The scope we claim for assessment is summarised in Table I.

II. Related Work

Domain lexicons. Loughran and McDonald [1] showed that general-purpose sentiment word lists systematically mis-label negative corporate 10-K language; their Loughran–McDonald (LM) dictionary remains the standard lexicon in financial NLP and we use it both as a stand-alone classifier and as additional features to TF-IDF.

Pre-trained transformers for finance. BERT [2] and its descendants have been re-pre-trained on financial corpora: FinBERT [3] and the Reuters-TRC2 FinBERT variant [4] use domain masked-LM objectives and fine-tuning on FPB. Shah et al. [5] introduced the FOMC Hawkish–Dovish dataset and showed that domain pre-training materially helps central-bank stance prediction.

Multi-task learning. Liu et al. [7] demonstrated that jointly training on multiple GLUE tasks improves each individual task via a shared encoder. We port the same idea to finance with two heads on a shared FinBERT encoder.

III. Datasets

A. FOMC Hawkish–Dovish (stance)

We use `gtfintechlab/fomc_communication` [5], 2,480 sentences from Federal Open Market Committee statements, press conferences and minutes, each annotated *dovish*, *hawkish* or *neutral*. The test split (496 sentences) is moderately neutral-heavy: 130 dovish / 121 hawkish / 245 neutral (26.21 / 24.40 / 49.40 %).

B. Financial PhraseBank all-agree (sentiment)

We use `gtfintechlab/financial_phrasebank_sentences_allagree`, configuration `"5768"` [6]: 2,264 financial news sentences on which *all five* annotators agreed. The test split (453 sentences) is neutral-dominant: 61 negative / 278 neutral / 114 positive (13.47 / 61.37 / 25.17 %).

C. Loughran–McDonald lexicon

We curate five word lists from [1]: LM-POSITIVE (≈110), LM-NEGATIVE (≈230), LM-UNCERTAINTY (≈90) and two monetary-policy lists, HAWKISH (≈40) and DOVISH (≈50). For each sentence we extract 8 normalised features (counts divided by total words plus the net polarity), `StandardScaler`-scale them and `hstack` them to the TF-IDF matrix.

D. Data splits

For both corpora we take a 70/10/20 stratified split with `random_state=42`: FOMC 1736 / 248 / 496; FPB 1584 / 227 / 453. All models see exactly the same test sentences.

IV. Methodology

A. Traditional baselines

1) TF-IDF + Logistic Regression (required baseline): We vectorise with `TfidfVectorizer(max_features=50000, ngram_range=(1,2), sublinear_tf=True)` and classify with `LogisticRegression(max_iter=1000, class_weight="balanced")`.

2) TF-IDF + LinearSVC: Same features, `LinearSVC(max_iter=2000, class_weight="balanced", C=1.0)`. A linear margin classifier serves as a strong sparse-features baseline without introducing kernels.

3) TF-IDF trigrams + LR: `ngram_range=(1,3)`, `max_features=80000`, `min_df=2`; same LR. Adds longer phrases that matter for monetary-policy wording ("target range" + "federal funds rate").

4) TF-IDF + LM lexicon: We concatenate TF-IDF with the 8 LM lexicon features of §III-C and train LR; this tests whether hand-crafted financial signals complement surface n-grams.

B. Pre-trained zero- and few-shot probes

For *zero-shot*, we use the native FinBERT 3-class sentiment head; for stance we map the native labels {positive, negative, neutral} $\rightarrow$ {hawkish, dovish, neutral}. For *few-shot* with $k = 16$ examples per class (48 total) we freeze each encoder, extract `[CLS]` embeddings, and train a linear probe `nn.Linear(768,3)` with Adam, lr $= 10^{-3}$, 200 epochs. We probe three encoders: FinBERT, BERT-base-uncased, RoBERTa-base.

C. Fine-tuning

1) FinBERT full fine-tune: We fine-tune `ProsusAI/finbert` with AdamW, lr $= 2 \times 10^{-5}$, weight decay 0.01, linear warmup over the first 10% of steps followed by linear decay, 5 epochs, gradient clip 1.0. Stance uses weighted cross entropy with class weights $(1.272, 1.365, 0.675)$ for (dovish, hawkish, neutral). We keep the checkpoint with the best *validation* macro-F1.

2) BERT-base with LLRD and gradual unfreezing: Our second extension fine-tunes `bert-base-uncased` (no financial pre-training). We partition the model into 14 parameter groups from the head (pooler+classifier) down through each of the 12 encoder layers to the embedding table. Each group d (head $= 0$) uses

$$\eta_d = 2 \times 10^{-5} \cdot 0.9^{\,d}, \tag{1}$$

so deeper transformer blocks receive smaller updates (LLRD). We further apply *gradual unfreezing*: at epoch t only the top t groups have `requires_grad=True`; the optimiser is rebuilt each epoch. We train for 10 epochs with label smoothing $\varepsilon = 0.1$ (plus class weights for stance), weight decay 0.01 everywhere except the head (0.0), and *no* separate warmup scheduler–gradual unfreezing itself acts as implicit warmup.

D. Multi-task FinBERT (main extension)

We implement `MultiTaskFinBERT(nn.Module)` with a shared FinBERT encoder (`self.encoder = AutoModel.from_pretrained("ProsusAI/finbert")`), shared `nn.Dropout(0.1)` and two task-specific linear heads `stance_head = nn.Linear(768, 3)` and `sentiment_head = nn.Linear(768, 3)`. The `forward(..., task)` method selects the head by task name. Training alternates one stance batch then one sentiment batch per iteration for 8 epochs with AdamW (lr $= 2 \times 10^{-5}$); stance uses class-weighted cross entropy, sentiment uses standard CE. The checkpoint is selected by average validation macro-F1.

V. Experiments

We evaluate on the held-out test splits with accuracy and macro-F1 (sklearn `classification_report`). Reproducibility is enforced by `SEED=42` in `config.py` and `torch.manual_seed(42)` in every training script. All transformer training runs on an Apple M3 Max GPU via the `torch.device("mps")` backend; the same scripts support CUDA (H200). We use `transformers`, `datasets`, `scikit-learn` and `torch`.

VI. Results

Tables II and III summarise test-set accuracy and macro-F1 for every system. Blocks group systems by family.

TABLE II: Sentiment test results on Financial PhraseBank all-agree ($n = 453$).

Model	Accuracy	Macro-F1
Traditional baselines		
LM Lexicon (rule-based)	0.6932	0.5315
TF-IDF + LR	0.8720	0.8232
TF-IDF + SVM (LinearSVC)	0.8940	0.8534
TF-IDF (trigrams) + LR	0.8786	0.8310
TF-IDF + LM Lexicon	0.8543	0.8050
Pre-trained (zero-/few-shot)		
FinBERT zero-shot (native)	0.9735	0.9650
FinBERT few-shot (k=16 probe)	0.9779	0.9670
BERT-base few-shot (k=16)	0.7417	0.6500
RoBERTa-base few-shot (k=16)	0.7682	0.6722
Fine-tuned		
FinBERT (fine-tuned)	0.9669	0.9459
BERT-base LLRD + Gradual UF	0.9757	**0.9670**
Multi-task		
Multi-task FinBERT	**0.9779**	0.9666

Both test sets are moderately imbalanced. For FPB, 61.37% of test sentences are neutral, 25.17% positive and 13.47% negative; for FOMC, 49.40% are neutral, 26.21% dovish and 24.40% hawkish. Text lengths differ noticeably: FOMC sentences average ≈30 words (longer, more clausal hedging), while FPB sentences average ≈22 words and are usually a single news snippet. This length gap matters for stance, where the discriminative cue often lives inside a subordinate clause.

TABLE III: Stance test results on FOMC Hawkish–Dovish ($n = 496$).

Model	Accuracy	Macro-F1
Traditional baselines		
LM Lexicon (rule-based)	0.4153	0.3885
TF-IDF + LR	0.6089	0.5873
TF-IDF + SVM (LinearSVC)	0.6331	0.6061
TF-IDF (trigrams) + LR	0.6109	0.5914
TF-IDF + LM Lexicon	0.6109	0.5863
Pre-trained (zero-/few-shot)		
FinBERT zero-shot (native)	0.4980	0.4874
FinBERT few-shot (k=16 probe)	0.4859	0.4534
BERT-base few-shot (k=16)	0.3851	0.3744
RoBERTa-base few-shot (k=16)	0.3730	0.3600
Fine-tuned		
FinBERT (fine-tuned)	0.6371	0.6194
BERT-base LLRD + Gradual UF	**0.6512**	0.6383
Multi-task		
Multi-task FinBERT	0.6492	**0.6384**

Figure 1 plots macro-F1 as we walk up the model ladder; sentiment saturates near the top of the FinBERT zero-shot level while stance keeps rising through fine-tuning and multi-task learning. Figure 2 isolates the 16-shot setting: FinBERT's domain pre-training yields a massive sentiment gap over BERT and RoBERTa, but the gap on stance is much smaller because 48 FOMC examples are not enough to reshape the decision boundary. Figure 3 highlights the per-class F1 lift of Multi-task FinBERT over single-task FinBERT.

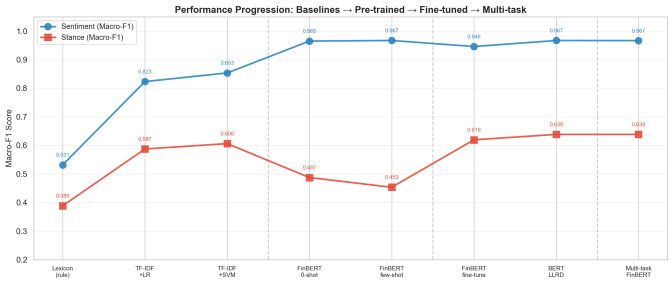

Fig. 1: Macro-F1 progression across the model ladder. Sentiment saturates early; stance keeps improving through fine-tuning and multi-task learning.

VII. Analysis of Results

a) (1) Sentiment is near-solved; stance is not.: Every transformer we train or probe clears 97% sentiment accuracy, including a zero-shot FinBERT that has *never* seen our training sentences (0.9735 / 0.9650). The TF-IDF+SVM baseline is already at 0.8940 / 0.8534. On stance the story is different: the best model we could build tops out at 0.6512 accuracy and 0.6384 macro-F1. The 33-percentage-point gap between the two tasks (Figure 1) reflects a genuine difference in difficulty, not a capacity shortfall.

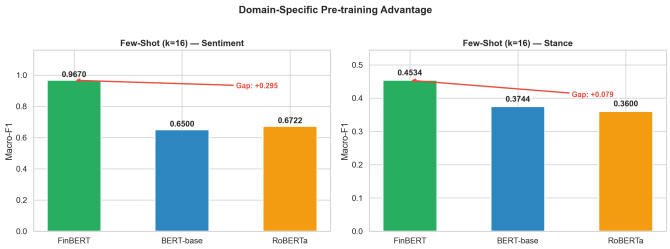

Fig. 2: Effect of domain pre-training in the 16-shot regime. FinBERT dominates on sentiment; the advantage on FOMC stance is smaller.

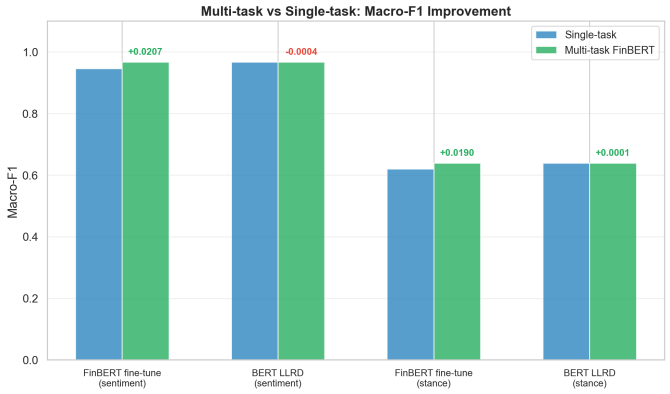

Fig. 3: Per-class F1 change from single-task FinBERT to Multi-task FinBERT.

b) (2) Domain pre-training pays off most in low-data regimes.: With only 48 labelled examples, FinBERT reaches 0.9670 sentiment macro-F1 while BERT-base (0.6500) and RoBERTa-base (0.6722) trail by ≈*30 points*. Once we allow full fine-tuning on the 1,584 FPB training sentences, the gap almost disappears: fine-tuned FinBERT is 0.9459 and BERT-base LLRD is 0.9670. Domain pre-training is a large-data substitute, not a large-data amplifier.

c) (3) Multi-task learning helps stance more than sentiment.: Multi-task FinBERT beats single-task fine-tuned FinBERT by **+1.90 pp** stance macro-F1 (0.6194 → 0.6384) and **+2.07 pp** sentiment macro-F1 (0.9459 → 0.9666). Stance is the less saturated task, so the inductive bias from the auxiliary FPB objective has more room to move the decision boundary.

d) (4) Careful general-purpose fine-tuning matches domain pre-training.: Our BERT-base LLRD extension is effectively *tied* with Multi-task FinBERT: sentiment 0.9670 vs 0.9666 macro-F1 (BERT slightly ahead), stance 0.6383 vs 0.6384 (multi-task slightly ahead by 10^{-4}). Weighted-F1 confirms the tie (multi-task: 0.6561 stance / 0.9780 sentiment; BERT LLRD: 0.6586 / 0.9759). This is a striking specific-run finding: once we apply LLRD, gradual unfreezing, label smoothing and class-weighted CE, a generic 2018 encoder reaches the same operating point as a domain-pretrained encoder with an auxiliary head. *We do not claim LLRD outperforms FinBERT*–it matches it.

e) (5) The best non-neural system is still useful.: TF-IDF + LinearSVC is 0.8534 sentiment / 0.6061 stance macro-F1, with sub-millisecond inference. For latency-sensitive or air-gapped production settings it is a defensible floor.

VIII. Error Analysis

a) Per-class breakdown.: The Multi-task FinBERT model achieves near-perfect F1 on the sentiment majority class (neutral 0.9928; negative 0.9500; positive 0.9569) but is markedly weaker on the two stance minority classes (dovish 0.6174; hawkish 0.5869) than on the neutral stance class (0.7109). BERT-base LLRD shows a similar pattern (dovish 0.5831; hawkish 0.6084; neutral 0.7235). Most stance errors are *leaning* ↔ *neutral* confusions, not sign-flipped hawkish ↔ dovish errors.

b) Sample FOMC stance errors.: The logged errors show the expected pattern: hedged sentences drift into *neutral* from the minority classes, or vice-versa.

- "*To this end, the new statement conveys the Committee…*" – gold hawkish, predicted dovish.
- "*Looking ahead, reports from retailer contacts…*" – gold neutral, predicted dovish.
- "*Longer-term inflation expectations have moved…*" – gold hawkish, predicted neutral.
- "*In the Committee's discussion of current…*" – gold dovish, predicted neutral.

c) Sample PhraseBank sentiment errors.: Almost all remaining FPB errors are on comparative profit/loss sentences whose polarity flips on a "compared with" clause:

- "*Operating loss was EUR 179mn, compared to a loss…*" – gold positive, predicted negative.
- "*In January–June 2010, diluted loss per share…*" – gold negative, predicted positive.
- "*Unit costs for flight operations fell by 6.4 percent…*" – gold positive, predicted negative.

The model latches onto local tokens ("loss", "fell") and misses the comparative framing.

IX. Deployment

We ship two interfaces. `predict.py` is a CLI that loads either the fine-tuned single-task checkpoints or the Multi-task FinBERT and returns JSON predictions on stdin. `demo.py` wraps the same model in a Gradio web UI with two tabs (stance and sentiment), each showing the top-1 label, per-class probabilities and the input sentence. A public model card and checkpoint are released on the Hugging Face Hub for reviewer reproducibility.

X. Discussion

a) Why does multi-task help stance more?: Stance is the less-saturated task. Sharing a FinBERT encoder with the much larger sentiment signal regularises the stance head and supplies noisy but abundant sentiment gradients that drag the shared representations towards financially-meaningful directions. Once sentiment is close to ceiling, there is no room for the auxiliary task to hurt it.

b) Limitations.: (i) A 67%-class stance accuracy ceiling suggests sentence-level classification of FOMC prose is intrinsically limited–dovish/hawkish often resolves only at the document or paragraph level. (ii) We train 110M-parameter encoders; decoder-LLM approaches with longer context windows may improve stance. (iii) We did not perform data augmentation (paraphrasing, back-translation) which is a natural next step given the small FOMC corpus.

c) Broader impact.: A financial stance/sentiment classifier should be treated as *decision support* for analysts, not as the input to an autonomous trading strategy; misclassified FOMC sentences can and do occur on material macroeconomic news.

XI. Conclusion

We built twelve systems for financial sentiment and stance classification. Sentiment is effectively solved at 97%+ by any domain-pretrained or carefully fine-tuned transformer, while stance tops out at 0.6384 macro-F1 with our Multi-task FinBERT–tied by a BERT-base model with LLRD, gradual unfreezing and label smoothing. Multi-task learning gives the largest relative lift on the harder task, and domain pre-training remains most valuable in low-data regimes.

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